

Simulation of water permeation in a CNT membrane

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Question 1

Analyse the behaviour of water inside the nanotubes during the equilibrium state. In particular use the python scripts to compute: (a) the number of water molecules inside the nanotubes as a function of time, (b) the number of permeations in each direction (from +z to -z and vice versa) and (c) the flow of water molecules. How do you interpret these results?

a. Number of water molecules inside the nanotube in function of time

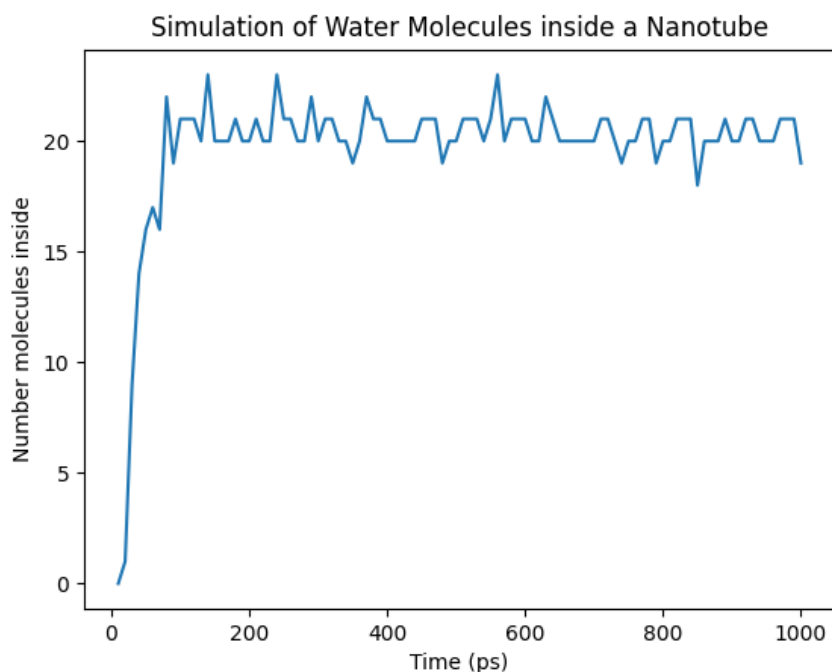


Figure 1: Number of water molecules inside the nanotubes as a function of time

The Figure 1 shows the number of water molecules inside the nanotubes as a function of time. As can be observed, there is an initial stage between 0 and approximately 80 ps in which the number of molecules increases sharply. This behaviour is associated with the initial condition of the nanotube, which is empty at the beginning and gradually fills with water molecules from the surrounding environment.

After this initial increase, the curve reaches an equilibrium stage from approximately 80 ps to 1000 ps. In this regime, the number of molecules stabilises around $N \approx 20$, exhibiting fluctuations between approximately 18 and 23 molecules.

In conclusion, the figure illustrates how the system reaches a stationary filling state. The observed fluctuations are associated with the continuous dynamic exchange of water molecules with the environment. On average, for each molecule that leaves the nanotube, another one enters. During the equilibrium stage, the nanotube is fully occupied, forming a continuous one-dimensional chain of water molecules inside it. The described behaviour is shown in Figure ??

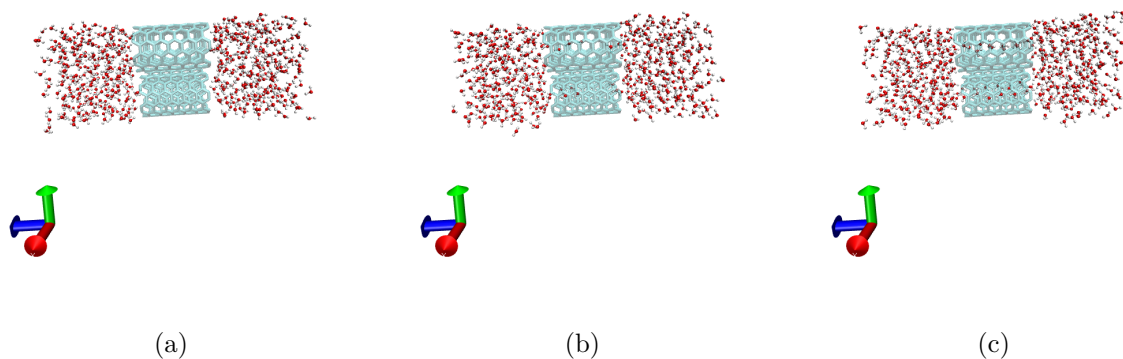


Figure 2: Evolution of the number of water molecules inside the nanotube

b. Number of permeations in each direction

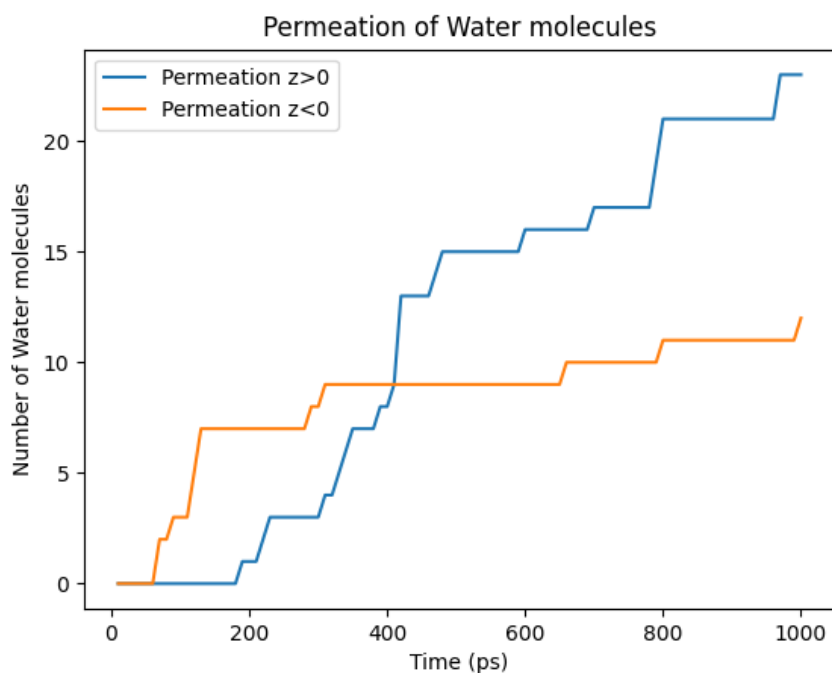


Figure 3: Number of permeations in each direction

Figure 3 shows the permeation in each direction of the nanotube. The curve corresponding to the +z direction (blue) has no values at the beginning, but starts to increase at approximately 200 ps. It then rises sharply in a step-like manner, reaching values above 20 molecules.

On the other hand, the curve corresponding to the -z direction (orange) is higher at the beginning, but after 200 ps it remains relatively stable, reaching a maximum value of around 10 water molecules.

The code calculates cycles of complete permeation events. In other words, a molecule is counted as permeated only if it enters the nanotube from one side, crosses it, and exits through the opposite side.

As shown in the figure, the curve corresponding to the +z direction reaches higher values than the curve associated with the -z direction. This indicates the presence of a preferential flux in the +z direction. Consequently, the system does not exhibit symmetric equilibrium, and there is a net flux in one direction.

c. Flow of water molecules

Figure 4 represents the accumulated net flux. Accordingly, the curve represents the net flux accumulated over time: when the curve increases, there is a net flux in the +z direction, whereas a decreasing trend indicates a net flux in the -z direction. Conversely, if the curve remains approximately constant, the system is in a state of dynamic equilibrium, with no net transport between both sides of the nanotube.

At the beginning, it can be observed that there is an initial flux in the -z direction, as the curve reaches negative values. Following this stage, an intermediate regime appears between approximately 200 and 500 ps. In this range, the curve increases steadily and crosses into positive values. This behaviour indicates a change in the transport regime, after which a net flux in the +z direction is established.

Finally, from approximately 500 ps to 1000 ps, the curve remains positive with noticeable fluctuations. This behaviour represents a steady-state regime characterized by a net flux in the +z direction, combined with a dynamic exchange of molecules between both sides of the nanotube.

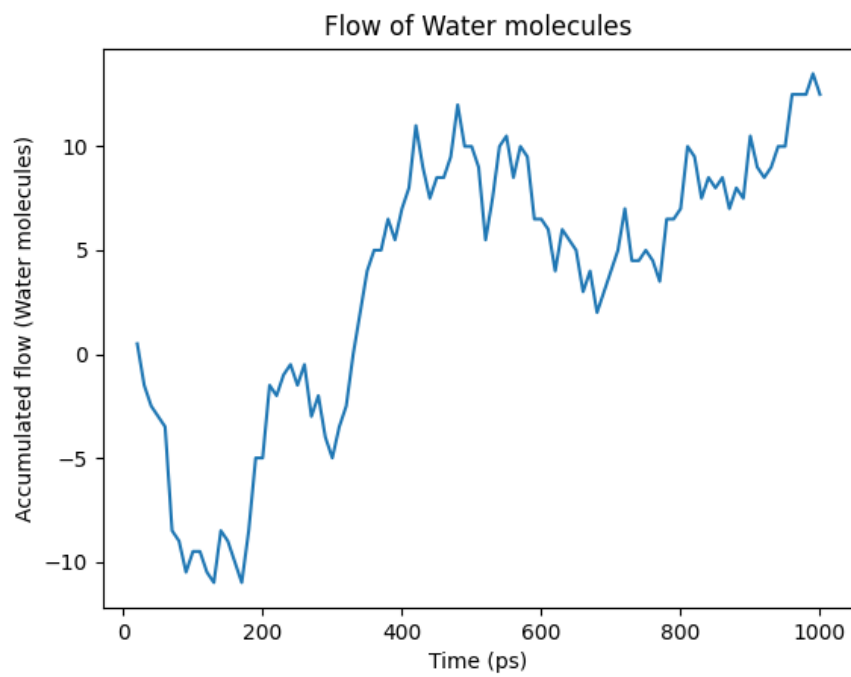


Figure 4: Flow of water molecules